

Relation Algebra

§15. Dynamic Programming

§15.1. Theory

$h : FB \rightarrow B$ a map, $T : FA \rightarrow A$ an F-algebra, $R : B \rightarrow B$.

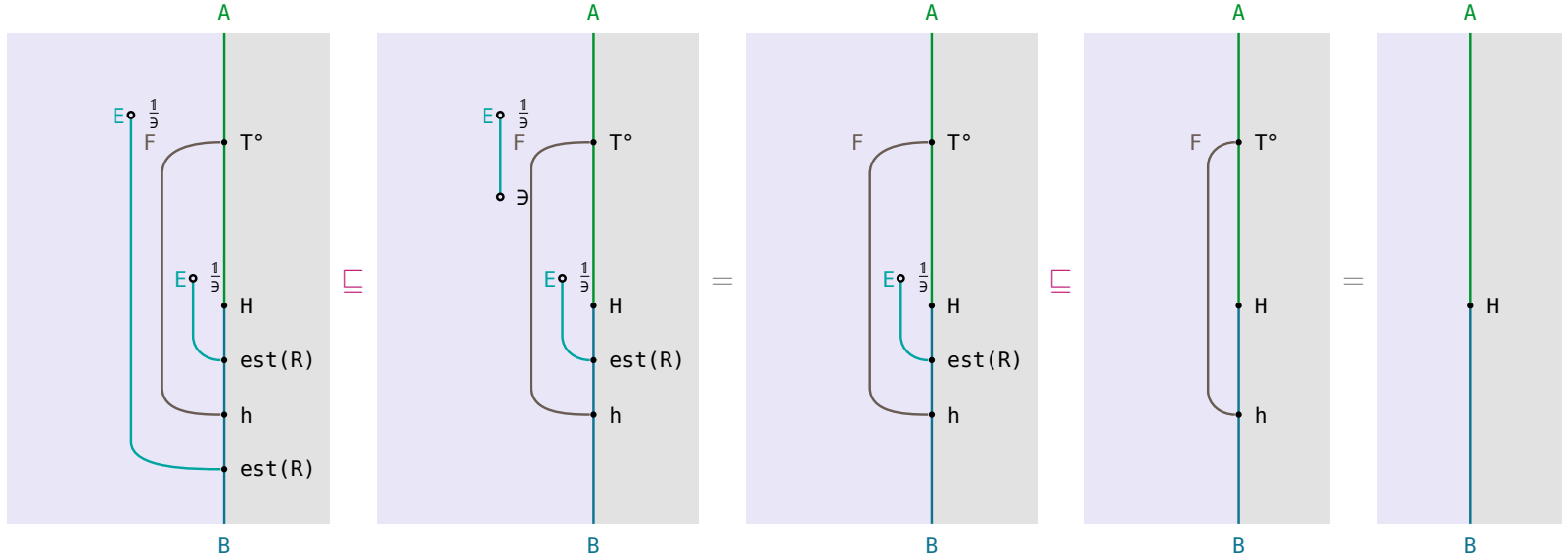
(15.1a)

$H \triangleq (T)^\circ \circ (h) : A \rightarrow B$, $M \triangleq \frac{H}{\exists} \text{ est}(R)$ the problem to be solved, $(\mu X : G(X))$ as in $\triangleq \mu$.

$$T^\circ F(H)h = H \implies \frac{T^\circ}{\exists} P(F(\frac{H}{\exists} \text{ est}(R))h) \text{ est}(R) \sqsubseteq H$$

(15.1b)

taking the input apart every way T allows $(\frac{T^\circ}{\exists})$, solving each part by M and keeping an optimum $(P(F(M)h) \text{ est}(R))$ returns only what H returns



$$\begin{aligned}
 & A \xrightarrow{\frac{1}{\exists}} EA \xrightarrow{E(T^\circ)} EFA \xrightarrow{P(F(\frac{H}{\exists} \text{ est}(R))h)} EB \xrightarrow{\text{est}(R)} B \\
 \sqsubseteq & A \xrightarrow{\frac{1}{\exists}} EA \xrightarrow{E(T^\circ)} EFA \xrightarrow{\exists} FA \xrightarrow{F(\frac{H}{\exists} \text{ est}(R))} FB \xrightarrow{h} B \\
 = & A \xrightarrow{T} FA \xrightarrow{F(\frac{H}{\exists} \text{ est}(R))} FB \xrightarrow{h} B \\
 \sqsubseteq & A \xrightarrow{T} FA \xrightarrow{F(H)} FB \xrightarrow{h} B \\
 = & A \xrightarrow{H} B
 \end{aligned}$$

$$(9.4) P(X) \text{ est}(R) \sqsubseteq \exists X \text{ at } X \triangleq F(\frac{H}{\exists} \text{ est}(R))h \text{ — est-710}$$

$$\frac{T^\circ}{\exists} \exists = T^\circ \text{ — pow-laws}$$

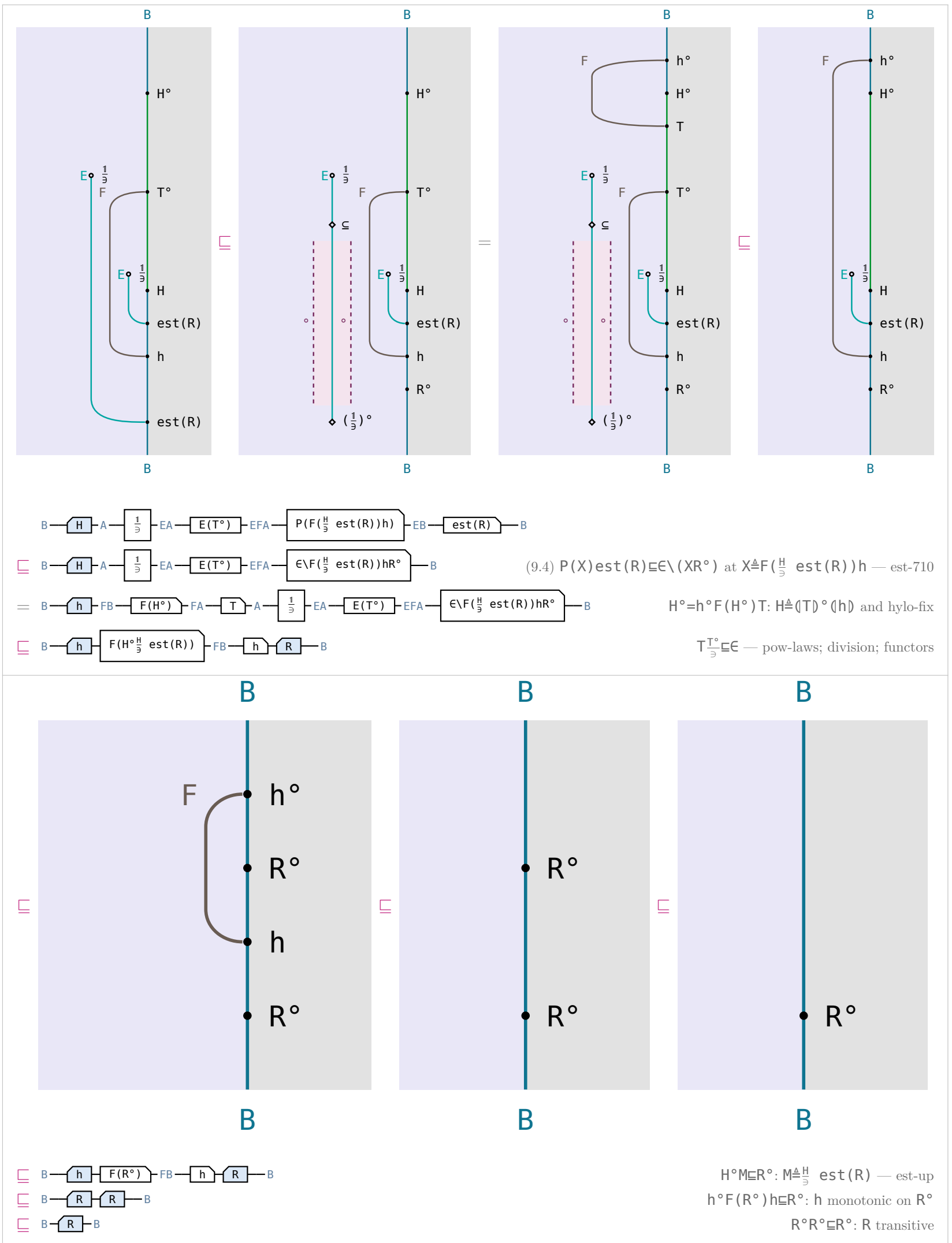
$$M \triangleq \frac{H}{\exists} \text{ est}(R) \sqsubseteq \frac{H}{\exists} \exists = H \text{ — est-up}$$

$$T^\circ F(H)h = H: H \triangleq (T)^\circ \circ (h) \text{ and hylo-fix}$$

$$\text{PreservesRecip}(F) \wedge \text{Map}(h) \wedge \text{MonotonicAlg}(h, R^\circ) \wedge R^\circ R^\circ \sqsubseteq R^\circ \wedge T^\circ F(H)h = H \implies H^\circ \frac{T^\circ}{\exists} P(F(\frac{H}{\exists} \text{ est}(R))h) \text{ est}(R) \sqsubseteq R^\circ$$

(15.1c)

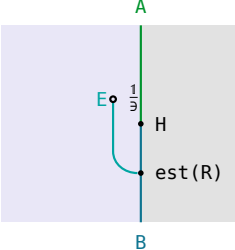
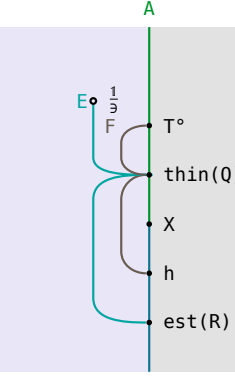
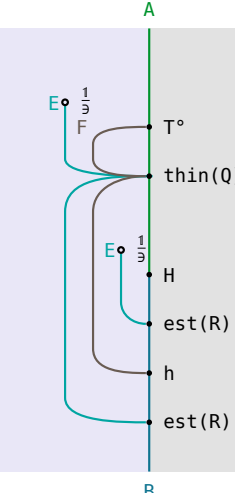
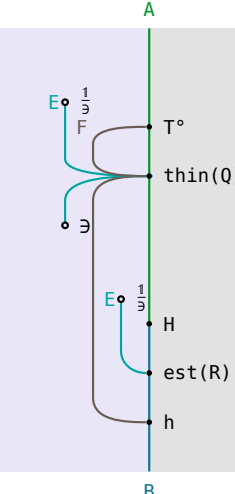
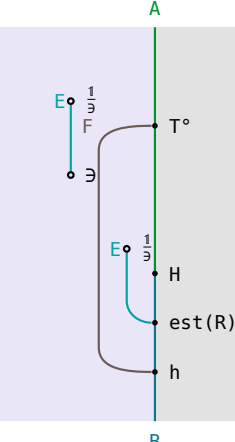
for every b that H returns from an input, the step $\frac{T^\circ}{\exists} P(F(M)h) \text{ est}(R)$ returns from that input only b' with R relating b' to b



PreservesRecip(F) $\wedge$ Map(h) $\wedge$ MonotonicAlg(h, R°) $\wedge$ $R^\circ R^\circ \sqsubseteq R^\circ \wedge QF(H)h \sqsubseteq F(H)hR \implies (\mu X : \frac{T^\circ}{3} \text{ thin}(Q)P(F(X)h) \text{ est}(R)) \sqsubseteq \frac{H}{3} \text{ est}(R)$

an optimum over everything H returns is reached by taking the input apart every way T allows, dropping the parts that can never win, solving each of the rest and keeping one optimum

(15.1d)

circuit	Hinze–Marsden
$A \rightarrow \boxed{\frac{1}{\exists}} \rightarrow EA \rightarrow \boxed{E(H)} \rightarrow EB \rightarrow \boxed{\text{est}(R)} \rightarrow B$ the problem to be solved, $H \triangleq (T)^\circ (h) \rightarrow \triangleq H$	
$A \rightarrow \boxed{\frac{1}{\exists}} \rightarrow EA \rightarrow \boxed{E(T^\circ)} \rightarrow \boxed{\text{thin}(Q)} \rightarrow EFA \rightarrow \boxed{P(F(X)h)} \rightarrow EB \rightarrow \boxed{\text{est}(R)} \rightarrow B$ $\sqsubseteq$ h monotonic on R and Q a preorder with $QF(H)h \in F(H)hR$; $\text{thin}(Q)$ as in thin-laws. Theorem 9.1 is this with the thinning step dropped — $1 \sqsubseteq \text{thin}(Q)$. Knaster–Tarski leaves (9.1): the body at M is $\sqsubseteq M$; $M = Hn(H^\circ \setminus R^\circ)$ splits that into (9.2) and (9.3) below — est-up. The fixed point is unique and entire when T° followed by F's membership relation is inductive, $\frac{T^\circ}{\exists}$ finite and non-empty, R connected	
circuit	Hinze–Marsden
$A \rightarrow \boxed{\frac{1}{\exists}} \rightarrow EA \rightarrow \boxed{E(T^\circ)} \rightarrow \boxed{\text{thin}(Q)} \rightarrow EFA \rightarrow \boxed{P(F(\frac{H}{\exists} \text{est}(R))h)} \rightarrow EB \rightarrow \boxed{\text{est}(R)} \rightarrow B$ (9.2)	
$A \rightarrow \boxed{\frac{1}{\exists}} \rightarrow EA \rightarrow \boxed{E(T^\circ)} \rightarrow \boxed{\text{thin}(Q)} \rightarrow EFA \rightarrow \boxed{\exists} \rightarrow FA \rightarrow \boxed{F(\frac{H}{\exists} \text{est}(R))} \rightarrow FB \rightarrow \boxed{h} \rightarrow B$ (9.4) $P(X)\text{est}(R) \sqsubseteq \exists X$ at $X \triangleq F(\frac{H}{\exists} \text{est}(R))h$ — est-710	
$A \rightarrow \boxed{\frac{1}{\exists}} \rightarrow EA \rightarrow \boxed{E(T^\circ)} \rightarrow EFA \rightarrow \boxed{\exists} \rightarrow FA \rightarrow \boxed{F(\frac{H}{\exists} \text{est}(R))} \rightarrow FB \rightarrow \boxed{h} \rightarrow B$ $\text{thin}(Q) \exists \sqsubseteq \exists$ — thin-laws	

(15.1e)

circuit	Hinze–Marsden
$ \begin{aligned} &= A \text{---} \boxed{T} \text{---} FA \text{---} \boxed{F(\frac{H}{\exists} \text{ est}(R))} \text{---} FB \text{---} \boxed{h} \text{---} B \\ &= \frac{T^\circ}{\exists} \ni T^\circ \text{ --- pow-laws} \end{aligned} $	
$ \begin{aligned} &= A \text{---} \boxed{T} \text{---} FA \text{---} \boxed{F(H)} \text{---} FB \text{---} \boxed{h} \text{---} B \\ &= \frac{M \ni \frac{H}{\exists} \ni H \text{ --- est-up}}{M \ni \frac{H}{\exists} \ni H} \end{aligned} $	
$ \begin{aligned} &= A \text{---} \boxed{H} \text{---} B \\ &= T^\circ F(H) h = H: H \triangleq (T)^\circ (h) \text{ and hylo-fix} \end{aligned} $	

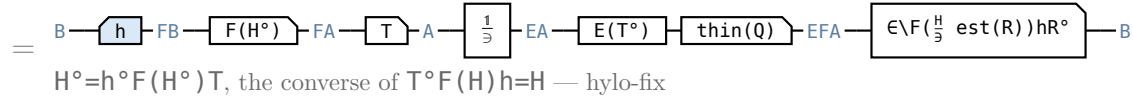
$$\text{PreservesRecip}(F) \wedge \text{Map}(h) \wedge \text{MonotonicAlg}(h, R^\circ) \wedge R^\circ R^\circ \subseteq R^\circ \wedge T^\circ F(H) h = H \wedge QF(H) h \subseteq F(H) h R \implies H^\circ \frac{T^\circ}{\exists} \text{ thin}(Q) P(F(\frac{H}{\exists} \text{ est}(R)) h) \text{ est}(R) \subseteq R^\circ$$

H° followed by the body at M is $\underline{E}R^\circ$: an answer of the body is never worse than an answer of H to the same input

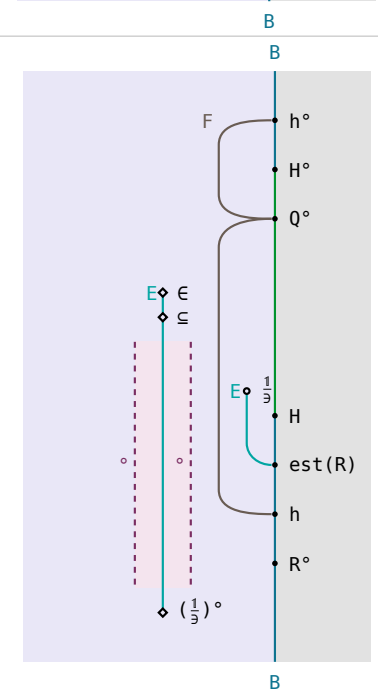
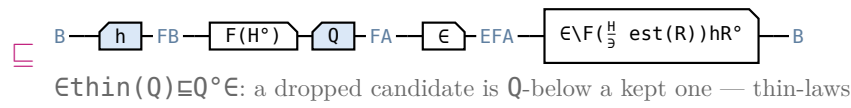
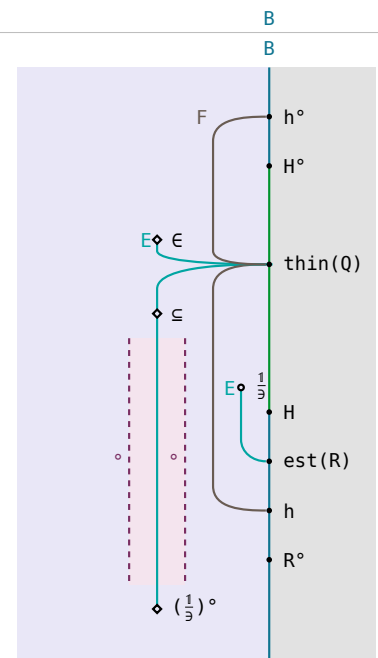
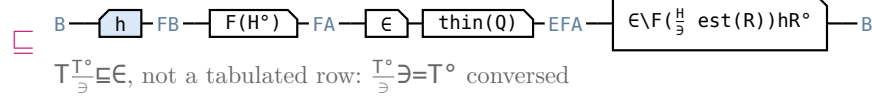
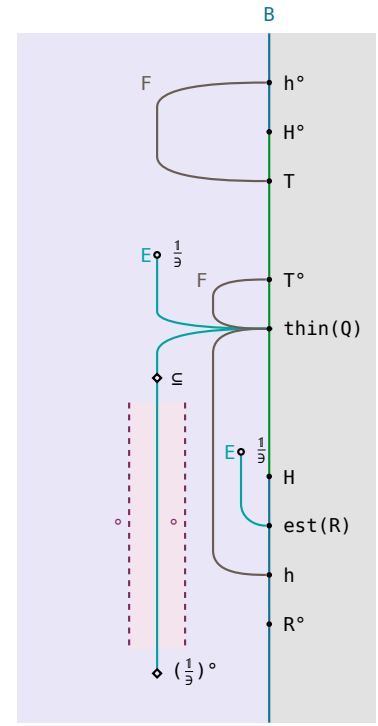
(15.1f)

circuit	Hinze–Marsden
$ \begin{aligned} &B \text{---} \boxed{H} \text{---} A \text{---} \boxed{\frac{1}{\exists}} \text{---} EA \text{---} \boxed{E(T^\circ)} \text{---} \boxed{\text{thin}(Q)} \text{---} EFA \text{---} \boxed{P(F(\frac{H}{\exists} \text{ est}(R)) h)} \text{---} EB \text{---} \boxed{\text{est}(R)} \text{---} B \end{aligned} $	
$ \begin{aligned} &= B \text{---} \boxed{H} \text{---} A \text{---} \boxed{\frac{1}{\exists}} \text{---} EA \text{---} \boxed{E(T^\circ)} \text{---} \boxed{\text{thin}(Q)} \text{---} EFA \text{---} \boxed{E \setminus F(\frac{H}{\exists} \text{ est}(R)) h R^\circ} \text{---} B \\ &(9.4) P(X) \text{ est}(R) \subseteq E \setminus (XR^\circ) \text{ at } X \triangleq F(\frac{H}{\exists} \text{ est}(R)) h \text{ --- est-710} \end{aligned} $	

circuit



Hinze–Marsden



circuit	Hinze–Marsden
A circuit diagram starting with input B. It passes through a box labeled h, followed by a wire labeled FB, a box labeled $F(H^\circ)$, a box labeled Q, a wire labeled FA, a box labeled E, a wire labeled EFA, a large box labeled $E \setminus F(\frac{H}{\exists} \text{est}(R))hR^\circ$, and finally output B.	The Hinze–Marsden diagram shows a vertical blue line representing the object B. Points from top to bottom are h°, H°, Q°, H, $\text{est}(R)$, h, and R°. A brown curve F connects h° and Q°. A teal curve E connects h° and $(\frac{1}{3})^\circ$. A pink shaded region is bounded by two vertical dashed lines.
A circuit diagram starting with input B. It passes through a box labeled h, followed by a wire labeled FB, a box labeled $F(H^\circ)$, a box labeled Q, a wire labeled FA, a box labeled $F(\frac{H}{\exists} \text{est}(R))$, a wire labeled FB, a box labeled h, a box labeled R, and finally output B. $E(E \setminus Y) \sqsubseteq Y$, not a tabulated row: division cancels	The Hinze–Marsden diagram shows a vertical blue line representing the object B. Points from top to bottom are h°, H°, Q°, H, $\text{est}(R)$, h, and R°. A brown curve F connects h° and Q°. A teal curve E connects h° and $(\frac{1}{3})$.
A circuit diagram starting with input B. It passes through a box labeled R, a box labeled h, a wire labeled FB, a box labeled $F(H^\circ)$, a wire labeled FA, a box labeled $F(\frac{H}{\exists} \text{est}(R))$, a wire labeled FB, a box labeled h, a box labeled R, and finally output B. $QF(H)h \sqsubseteq F(H)hR$ conversed — the hypothesis on Q	The Hinze–Marsden diagram shows a vertical blue line representing the object B. Points from top to bottom are R°, h°, H°, H, $\text{est}(R)$, h, and R°. A brown curve F connects h° and h. A teal curve E connects h° and $(\frac{1}{3})$.
A circuit diagram starting with input B. It passes through a box labeled R, a box labeled h, a box labeled $F(R^\circ)$, a wire labeled FB, a box labeled h, a box labeled R, and finally output B. $H^\circ M \sqsubseteq R^\circ$ under F — est-up	The Hinze–Marsden diagram shows a vertical blue line representing the object B. Points from top to bottom are R°, h°, R°, h, and R°. A brown curve F connects h° and h.
A circuit diagram starting with input B. It passes through three boxes labeled R in sequence, and finally output B. $h^\circ F(R^\circ)h \sqsubseteq R^\circ$: h monotonic on R, shunted — the hypothesis on h	The Hinze–Marsden diagram shows a vertical blue line representing the object B. Points from top to bottom are R°, R°, and R°.

circuit	Hinze–Marsden
$\begin{array}{c} B \text{---} \boxed{R} \text{---} B \\ \sqsubseteq \\ R \text{ transitive, twice} \end{array}$	

$$V_2 \quad V_1^\circ = \emptyset \implies \frac{[V_1, V_2]^\circ}{\exists} \text{thin}(Q_1 + Q_2) P([U_1, U_2]) \text{est}(R) = \text{ran}(V_1) \frac{V_1^\circ}{\exists} \text{thin}(Q_1) P(U_1) \text{est}(R) \cup \text{ran}(V_2) \frac{V_2^\circ}{\exists} \text{thin}(Q_2) P(U_2) \text{est}(R)$$

(15.1h)

when V_1 and V_2 have disjoint ranges, the body over $[V_1, V_2]$ runs the V_1 problem on inputs V_1 reaches and the V_2 problem on inputs V_2 reaches

circuit	Hinze–Marsden
$A \text{---} \boxed{\frac{1}{\exists}} \text{---} EA \text{---} E([V_1, V_2]^\circ) \text{---} \text{thin}(Q_1 + Q_2) \text{---} E(\alpha + \beta) \text{---} P([U_1, U_2]) \text{---} EB \text{---} \text{est}(R) \text{---} B$	
$\begin{array}{c} A \text{---} \text{ran}(V_1) \text{---} \boxed{\frac{1}{\exists}} \text{---} EA \text{---} E([V_1, V_2]^\circ) \text{---} \text{thin}(Q_1 + Q_2) \text{---} E(\alpha + \beta) \text{---} P([U_1, U_2]) \text{---} EB \text{---} \text{est}(R) \text{---} B \\ \cup \\ A \text{---} \text{ran}(V_2) \text{---} \boxed{\frac{1}{\exists}} \text{---} EA \text{---} E([V_1, V_2]^\circ) \text{---} \text{thin}(Q_1 + Q_2) \text{---} E(\alpha + \beta) \text{---} P([U_1, U_2]) \text{---} EB \text{---} \text{est}(R) \text{---} B \end{array}$ off $\text{ran } V_1 \cup \text{ran } V_2$ the set is empty and $\text{est}(R)$ of it is nothing; not a tabulated row	
$\begin{array}{c} A \text{---} \text{ran}(V_1) \text{---} \boxed{\frac{1}{\exists}} \text{---} EA \text{---} E(V_1^\circ) \text{---} E\alpha \text{---} P(l) \text{---} \text{thin}(Q_1 + Q_2) \text{---} E(\alpha + \beta) \text{---} P([U_1, U_2]) \text{---} EB \text{---} \text{est}(R) \text{---} B \\ \cup \\ A \text{---} \text{ran}(V_2) \text{---} \boxed{\frac{1}{\exists}} \text{---} EA \text{---} E(V_2^\circ) \text{---} E\beta \text{---} P(r) \text{---} \text{thin}(Q_1 + Q_2) \text{---} E(\alpha + \beta) \text{---} P([U_1, U_2]) \text{---} EB \text{---} \text{est}(R) \text{---} B \end{array}$ Exercise 9.5: on $\text{ran } V_1$, $\frac{[V_1, V_2]^\circ}{\exists} = \frac{V_1^\circ}{\exists} P(\text{in } l)$, as V_1, V_2 have disjoint ranges	
$\begin{array}{c} A \text{---} \text{ran}(V_1) \text{---} \boxed{\frac{1}{\exists}} \text{---} EA \text{---} E(V_1^\circ) \text{---} \text{thin}(Q_1) \text{---} E\alpha \text{---} P(l) \text{---} E(\alpha + \beta) \text{---} P([U_1, U_2]) \text{---} EB \text{---} \text{est}(R) \text{---} B \\ \cup \\ A \text{---} \text{ran}(V_2) \text{---} \boxed{\frac{1}{\exists}} \text{---} EA \text{---} E(V_2^\circ) \text{---} \text{thin}(Q_2) \text{---} E\beta \text{---} P(r) \text{---} E(\alpha + \beta) \text{---} P([U_1, U_2]) \text{---} EB \text{---} \text{est}(R) \text{---} B \end{array}$ Exercise 9.5: $P(\text{in } l) \text{thin}(Q_1 + Q_2) = \text{thin}(Q_1) P(\text{in } l)$	

circuit	Hinze–Marsden
$ \begin{array}{c} A \rightarrow \boxed{\text{ran}(V_1)} \rightarrow \boxed{\frac{1}{\beta}} \rightarrow EA \rightarrow \boxed{E(V_1^\circ)} \rightarrow \boxed{\text{thin}(Q_1)} \rightarrow E\alpha \rightarrow \boxed{P(U_1)} \rightarrow EB \rightarrow \boxed{\text{est}(R)} \rightarrow B \\ = \\ A \rightarrow \boxed{\text{ran}(V_2)} \rightarrow \boxed{\frac{1}{\beta}} \rightarrow EA \rightarrow \boxed{E(V_2^\circ)} \rightarrow \boxed{\text{thin}(Q_2)} \rightarrow E\beta \rightarrow \boxed{P(U_2)} \rightarrow EB \rightarrow \boxed{\text{est}(R)} \rightarrow B \\ P \text{ a relator and } \text{inl}[U_1, U_2] = U_1 \end{array} $	

$\text{Map}(\text{cost}) \wedge R = \text{cost} \text{ leq } \text{cost}^\circ \wedge h \text{cost} = F(\text{cost}) k \wedge F(\text{leq}) k \leq k \text{leq} \Rightarrow \text{MonotonicAlg}(h, R)$
 if R compares two values by comparing their **costs** under $\leq$, and h then **cost** equals $F(\text{cost})$ then a k monotonic on $\leq$, then h is monotonic on R

(15.1i)

circuit	Hinze–Marsden
$ \begin{array}{c} FA \rightarrow \boxed{F(R)} \rightarrow \boxed{h} \rightarrow A \rightarrow \boxed{\text{cost}} \rightarrow C \\ F(R)h \sqsubseteq hR \text{ iff } F(R)h \text{ cost} \sqsubseteq h \text{ cost} \leq: \text{definition of } R \text{ and shunting} \end{array} $	
$ \begin{array}{c} = \\ FA \rightarrow \boxed{F(R)} \rightarrow \boxed{F(\text{cost})} \rightarrow FC \rightarrow \boxed{k} \rightarrow C \\ \text{assumption } h \text{ cost} = F(\text{cost})k \end{array} $	
$ \begin{array}{c} \sqsubseteq \\ FA \rightarrow \boxed{F(\text{cost})} \rightarrow \boxed{F(\text{leq})} \rightarrow FC \rightarrow \boxed{k} \rightarrow C \\ R \text{ cost} \sqsubseteq \text{cost} \leq, \text{ as } \text{cost} \text{ is a map; functors} \end{array} $	
$ \begin{array}{c} \sqsubseteq \\ FA \rightarrow \boxed{F(\text{cost})} \rightarrow FC \rightarrow \boxed{k} \rightarrow \boxed{\text{leq}} \rightarrow C \\ \text{assumption } F(\leq) k \leq k \leq: k \text{ monotonic on } \leq \end{array} $	
$ \begin{array}{c} = \\ FA \rightarrow \boxed{h} \rightarrow A \rightarrow \boxed{\text{cost}} \rightarrow \boxed{\text{leq}} \rightarrow C \\ \text{assumption } h \text{ cost} = F(\text{cost})k \end{array} $	

the condition	how it is discharged
$F(R)h \sqsubseteq hR$ Proposition 9.2, $R \triangleq \text{cost} \leq \text{cost}^\circ$, $h \text{ cost} = F(\text{cost})k$, $F(\leq) k \leq k \leq$	monotonicity when the cost is itself a fold with a step k monotonic on $\leq$
$F(Rn(H^\circ H))h \sqsubseteq hR$ Proposition 9.3, $R \triangleq \text{cost} \leq \text{cost}^\circ$, $h \text{ cost} = F(\langle \text{cost}, H^\circ \rangle)k$, $F(\leq \times 1) k \leq k \leq$, H° simple;	monotonicity in context : k may also read the input the part was built from
$QF(H)h \sqsubseteq F(H)hR$ at $Q \triangleq F(U, V)$ Proposition 9.4, U, V preorders, $F(U, R)h \sqsubseteq hR$, $VH \sqsubseteq HR$	both conditions at once, split along the two arguments of a bifunctor

(15.1j)

$$\text{Map}(\text{cost}) \wedge \text{Simple}(S) \wedge R = \text{cost} \text{ leq } \text{cost}^\circ \wedge \text{cost} = F(\langle \text{cost}, S \rangle) k \wedge F(\text{leq} \times 1) k \sqsubseteq k \text{ leq} \implies F(R \cap S S^\circ) h \sqsubseteq h R$$

improving each part by R within its S -context, then assembling by h , is below hR

circuit	Hinze–Marsden
$FA \text{---} \boxed{F(R \cap S S^\circ)} \text{---} \boxed{h} \text{---} A$	
$\sqsubseteq FA \text{---} \boxed{F(R \cap S S^\circ)} \text{---} \boxed{h} \text{---} A \text{---} \boxed{\text{cost}} \text{---} C \text{---} \boxed{\text{cost}} \text{---} A$ shunting: cost a map, so $1 \sqsubseteq \text{cost} \text{ cost}^\circ$ — triple-chains	
$= FA \text{---} \boxed{F(\langle \text{cost} \text{ leq}, S \rangle \langle \text{cost}, S \rangle^\circ)} \text{---} \boxed{h} \text{---} A \text{---} \boxed{\text{cost}} \text{---} C \text{---} \boxed{\text{cost}} \text{---} A$ products: $R \cap S S^\circ = \langle \text{cost} \text{ leq}, S \rangle \langle \text{cost}, S \rangle^\circ$ at $R = \text{cost} \text{ leq } \text{cost}^\circ \text{---} \triangleq \times$	
$= FA \text{---} \boxed{F(X)} \text{---} \boxed{F(\langle \text{cost}, S \rangle)} \text{---} F(C \times B) \text{---} \boxed{k} \text{---} C \text{---} \boxed{\text{cost}} \text{---} A$ assumption on cost : $h \text{ cost} = F(\langle \text{cost}, S \rangle) k$	
$\sqsubseteq FA \text{---} \boxed{F(\langle \text{cost} \text{ leq}, S \rangle)} \text{---} F(C \times B) \text{---} \boxed{k} \text{---} C \text{---} \boxed{\text{cost}} \text{---} A$ S simple, so $\langle \text{cost}, S \rangle^\circ \langle \text{cost}, S \rangle \sqsubseteq 1$	
$= FA \text{---} \boxed{F(\langle \text{cost}, S \rangle)} \text{---} \boxed{F(\text{leq} \times 1)} \text{---} F(C \times B) \text{---} \boxed{k} \text{---} C \text{---} \boxed{\text{cost}} \text{---} A$ products; functors: $\langle \text{cost} \text{ leq}, S \rangle = \langle \text{cost}, S \rangle (\text{leq} \times 1)$ — bdm-prod-laws, relator-laws	
$\sqsubseteq FA \text{---} \boxed{F(\langle \text{cost}, S \rangle)} \text{---} F(C \times B) \text{---} \boxed{k} \text{---} \boxed{\text{leq}} \text{---} C \text{---} \boxed{\text{cost}} \text{---} A$ assumption on k : $F(\text{leq} \times 1) k \sqsubseteq k \text{ leq}$	

circuit	Hinze–Marsden
$= \text{FA} \xrightarrow{\text{h}} \text{R} \xrightarrow{\text{A}}$ assumption on cost read backwards, then $\text{R} = \text{cost} \text{ leq } \text{cost}^\circ$	

$$\mathbf{G(U,R)h \in hR \wedge Vh \in HR \implies G(U,V)G(1,H)h \in G(1,H)hR}$$

(15.11)

thinning the parts by $\mathbf{U}$ in the first argument and by $\mathbf{V}$ in the second ($\mathbf{G(U,V)}$), then solving by $\mathbf{H}$ and assembling by $\mathbf{h}$, gives only what solving and assembling and then improving by $\mathbf{R}$ gives ($\mathbf{G(1,H)hR}$)

circuit	Hinze–Marsden
$\text{G e w} \xrightarrow{\text{G(U,V)}} \text{G(1,H)} \xrightarrow{\text{G e A}} \text{h} \xrightarrow{\text{A}}$	
$= \text{G e w} \xrightarrow{\text{G(U,VH)}} \text{G e A} \xrightarrow{\text{h}} \text{A}$ taking $\mathbf{Q \triangleq G(U,V)}$; bifunctors: $\mathbf{G(U,V)G(1,H) = G(U,VH)}$ — relator-laws	
$\sqsubseteq \text{G e w} \xrightarrow{\text{G(U,HR)}} \text{G e A} \xrightarrow{\text{h}} \text{A}$ assumption on $\mathbf{V}$: $\mathbf{Vh \in HR}$	
$= \text{G e w} \xrightarrow{\text{G(1,H)}} \text{G(U,R)} \xrightarrow{\text{G e A}} \text{h} \xrightarrow{\text{A}}$ bifunctors: $\mathbf{G(U,HR) = G(1,H)G(U,R)}$ — relator-laws	
$\sqsubseteq \text{G e w} \xrightarrow{\text{G(1,H)}} \text{G e A} \xrightarrow{\text{h}} \text{R} \xrightarrow{\text{A}}$ assumption on $\mathbf{h}$: $\mathbf{G(U,R)h \in hR}$	

§15.2. The string edit problem

$\text{Op} ::= \text{cpy Char} \mid \text{del Char} \mid \text{ins Char}, \text{F(A,B)} = 1 + (\text{A} \times \text{B}), \alpha \triangleq [\text{nil}, \text{cons}].$

(15.2a)

$\text{edit} \triangleq ([\text{base}, \text{step}]) : [\text{Op}] \rightarrow [\text{Char}] \times [\text{Char}]$, base returning $([], [])$, step $(\text{cpy } a, (xs, ys)) = ([a] \# xs, [a] \# ys)$, step $(\text{del } a, (xs, ys)) = ([a] \# xs, ys)$, step $(\text{ins } a, (xs, ys)) = (xs, [a] \# ys)$.

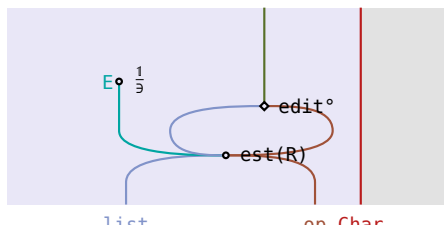
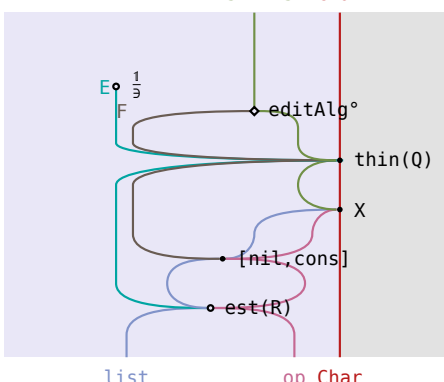
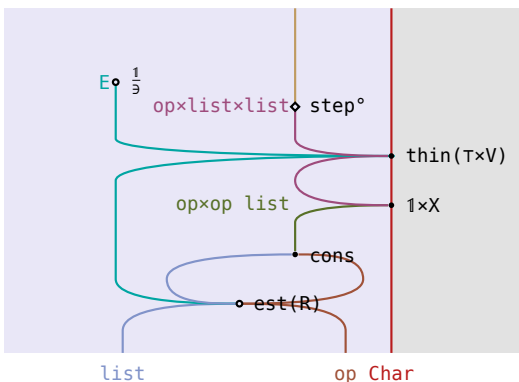
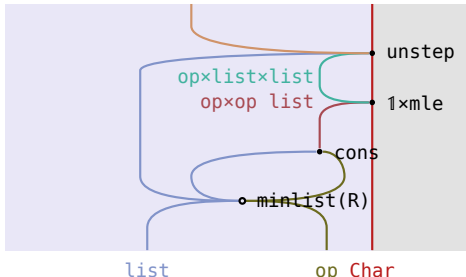
$\text{length} \triangleq ([\text{zero}, \pi_2 \text{ succ}])$, $\mathbf{R \triangleq \text{length} \leq \text{length}^\circ}$, $\mathbf{U \triangleq \top}$, $\mathbf{V \triangleq \text{suffix}^\circ \times \text{suffix}^\circ}$, $\mathbf{Q \triangleq 1 + (U \times V)}$, empty the coreflexive on (xs, ys) with both lists empty.

unstep implements $\frac{\text{step}^\circ}{\exists} \text{thin}(U \times V) : \text{unstep } ([a] \# xs, []) = [(\text{del } a, (xs, []))]$, $\text{unstep } ([], [b] \# ys) = [(\text{ins } b, ([], ys))]$, $\text{unstep } ([a] \# xs, [b] \# ys) = (a = b \rightarrow [(\text{cpy } a, (xs, ys))], [(\text{del } a, (xs, [b] \# ys)), (\text{ins } b, ([a] \# xs, ys))])$.

$(\mu X : \frac{\text{editAlg}^\circ}{\exists} \text{thin}(Q)P(\text{F(X)}[\text{nil}, \text{cons}])\text{est}(R)) \sqsubseteq \frac{\text{edit}^\circ}{\exists} \text{est}(R)$, $\text{mle} = (\text{empty} \rightarrow \text{nil}, \text{unstep list}((1 \times \text{mle})\text{cons}) \text{minlist}(R))$

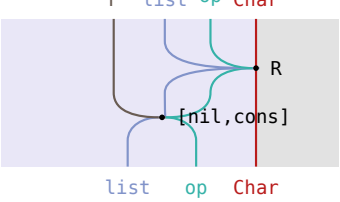
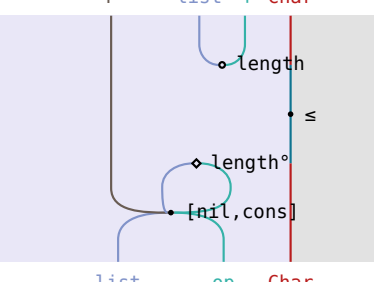
(15.2b)

a shortest edit sequence from which both strings can be reconstituted is one pass over the two of them, each step copying, deleting or inserting one character and the best sequence for what is left taken from the entries already computed

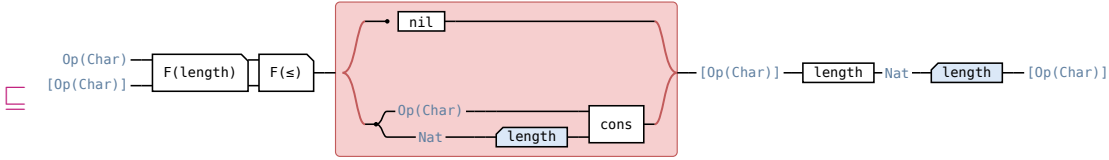
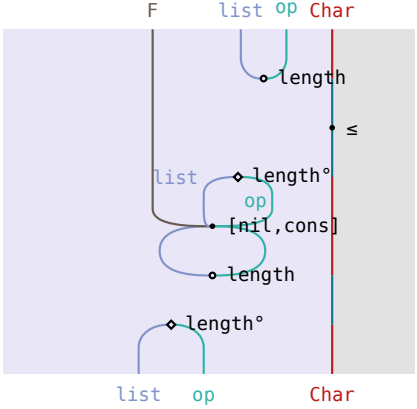
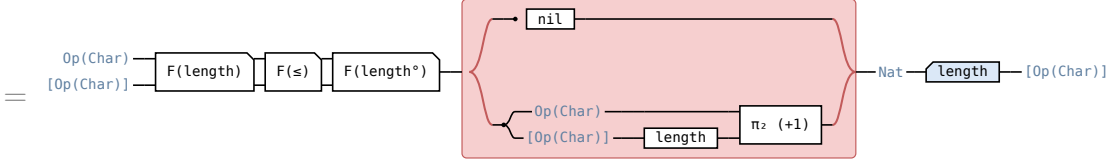
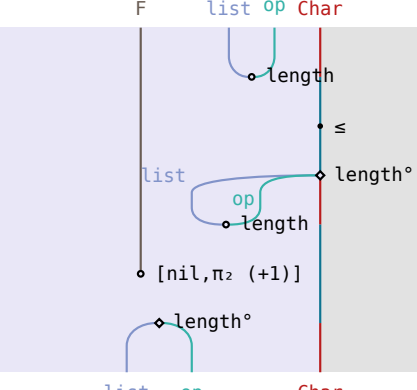
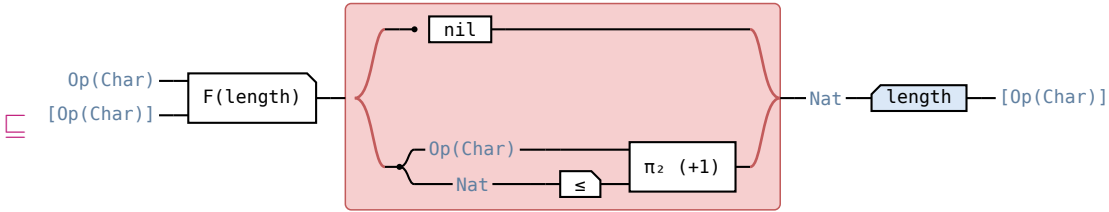
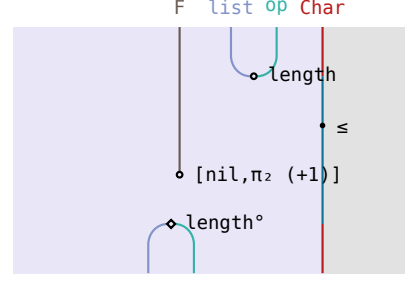
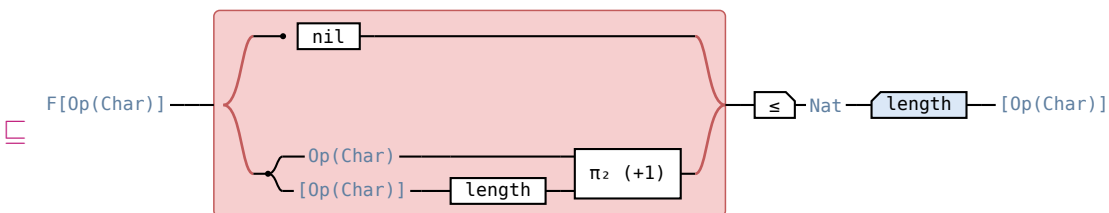
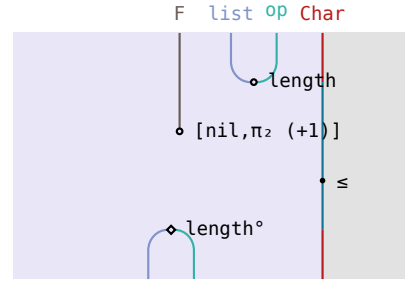
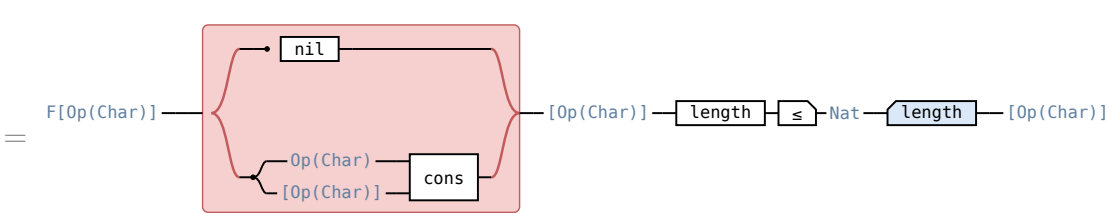
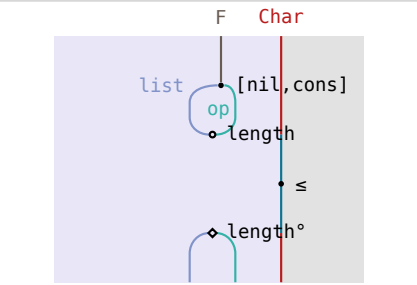
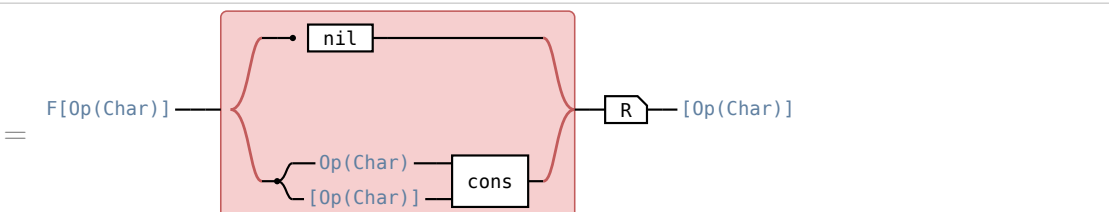
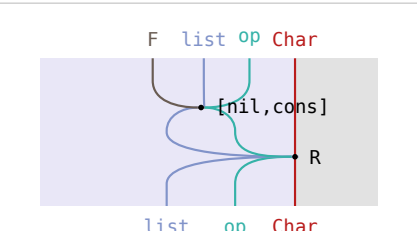
circuit	Hinze–Marsden
$ \begin{array}{c} [\text{Char}] \xrightarrow{\quad} \boxed{\frac{1}{3}} \xrightarrow{\quad} E[\text{Char}]^2 \xrightarrow{\quad} E(\text{edit}^\circ) \xrightarrow{\quad} E[\text{Op}(\text{Char})] \xrightarrow{\quad} \text{est}(\text{R}) \xrightarrow{\quad} [\text{Op}(\text{Char})] \\ \text{the specification} \quad \triangleq \quad \text{edit} \end{array} $	
$ \begin{array}{c} [\text{Char}] \xrightarrow{\quad} \boxed{\frac{1}{3}} \xrightarrow{\quad} E[\text{Char}]^2 \xrightarrow{\quad} E(\text{editAlg}^\circ) \xrightarrow{\quad} \text{thin}(Q) \xrightarrow{\quad} E[\text{Char}]^2 \xrightarrow{\quad} P(F(X)[\text{nil}, \text{cons}]) \xrightarrow{\quad} E[\text{Op}(\text{Char})] \xrightarrow{\quad} \text{est}(\text{R}) \xrightarrow{\quad} [\text{Op}(\text{Char})] \end{array} $ at $Q \triangleq 1 + (U \times V)$. Monotonicity $F(R) \alpha \sqsubseteq \alpha R$ is Proposition 9.2 at $\text{length} \triangleq ([\text{zero}, \pi_2 \text{succ}])$ with succ monotonic on $\leq$, so cons is monotonic on R. The thinning condition $QF(1, \text{edit}^\circ) \alpha \sqsubseteq F(1, \text{edit}^\circ) \alpha R$ over $F(\text{Op}, [\text{Char}] \times [\text{Char}])$ is Proposition 9.4 at $U \triangleq \tau \text{---} F(\tau, R) \alpha \sqsubseteq \alpha R$, left as an exercise, so any two operations may be compared — and $V \triangleq \text{suffix}^\circ \times \text{suffix}^\circ$, — $\text{edit} (1 \times \text{suffix}) \sqsubseteq R^\circ \text{edit}$, $\text{edit} (\text{suffix} \times 1) \sqsubseteq R^\circ \text{edit}$. $\text{suffix} = \text{tail}^*$ and $BA \sqsubseteq CB \implies BA^* \sqsubseteq C^*B$ cut those to one step each: drop the operation that produced the head, or weaken its cpy to a del, never lengthening the sequence	
$ \begin{array}{c} [\text{Char}] \xrightarrow{\quad} \boxed{\frac{1}{3}} \xrightarrow{\quad} E[\text{Char}]^2 \xrightarrow{\quad} E(\text{step}^\circ) \xrightarrow{\quad} \text{thin}(\tau \times V) \xrightarrow{\quad} E(\text{Op}(\text{Char}) \times [\text{Char}] \times [\text{Char}]) \xrightarrow{\quad} P((1 \times X) \text{cons}) \xrightarrow{\quad} E[\text{Op}(\text{Char})] \xrightarrow{\quad} \text{est}(\text{R}) \xrightarrow{\quad} [\text{Op}(\text{Char})] \end{array} $ $\equiv$ Proposition 9.1: base and step have disjoint ranges, so the fixed point splits into one branch per summand — empty, the coreflexive on (xs, ys) with both lists empty, is where base returns	
$ \begin{array}{c} [\text{Char}] \xrightarrow{\quad} \boxed{\text{unstep}} \xrightarrow{\quad} [\text{Op}(\text{Char}) \times [\text{Char}] \times [\text{Char}]] \xrightarrow{\quad} \text{list}((1 \times \text{mle}) \text{cons}) \xrightarrow{\quad} [[\text{Op}(\text{Char})]] \xrightarrow{\quad} \text{minlist}(\text{R}) \xrightarrow{\quad} [\text{Op}(\text{Char})] \end{array} $ $\equiv$ unstep implements $\frac{\text{step}^\circ}{3} \text{thin}(U \times V)$ — at most two decompositions survive, a copy beating a delete or an insert wherever it is available — and minlist(R) implements est(R). The same subproblem is solved many times over, so the running time is exponential in the two lengths	

$$F(R)[\text{nil}, \text{cons}] \sqsubseteq [\text{nil}, \text{cons}]R$$

comparing the tails by R and then putting an operation in front ($F(R)\alpha$) relates only sequences that R also relates once the operation is in front (αR)

circuit	Hinze–Marsden
$ \begin{array}{c} F[\text{Op}(\text{Char})] \xrightarrow{\quad} \boxed{\text{nil}} \xrightarrow{\quad} [\text{Op}(\text{Char})] \\ \text{Op}(\text{Char}) \xrightarrow{\quad} \boxed{\text{Op}(\text{Char})} \xrightarrow{\quad} R \xrightarrow{\quad} \boxed{\text{cons}} \xrightarrow{\quad} [\text{Op}(\text{Char})] \end{array} $	
$ \begin{array}{c} \text{Op}(\text{Char}) \xrightarrow{\quad} F(\text{length}) \xrightarrow{\quad} F(\leq) \xrightarrow{\quad} \boxed{\text{nil}} \xrightarrow{\quad} [\text{Op}(\text{Char})] \\ \text{Op}(\text{Char}) \xrightarrow{\quad} \boxed{\text{Op}(\text{Char})} \xrightarrow{\quad} \text{length} \xrightarrow{\quad} \boxed{\text{cons}} \xrightarrow{\quad} [\text{Op}(\text{Char})] \end{array} $ $R \triangleq \text{length} \leq \text{length}^\circ \triangleq \text{edit}$; F preserves composition	

(15.2c)

circuit	Hinze–Marsden
 length is a map, so entire: $1 \sqsubseteq \text{length} \text{ length}^\circ$	
 $\alpha \text{ length} = F(\text{length})[\text{zero}, \pi_2 \text{ succ}]: \text{length} \triangleq ([\text{zero}, \pi_2 \text{ succ}]) \triangleq \text{edit}$	
 length is a map, so simple, and F is monotonic: $F(\text{length}^\circ)F(\text{length}) \sqsubseteq F(1) = 1$	
 succ is monotonic on $\leq$: $F(\leq)[\text{zero}, \pi_2 \text{ succ}] \sqsubseteq [\text{zero}, \pi_2 \text{ succ}] \leq$	
 $F(\text{length})[\text{zero}, \pi_2 \text{ succ}] = \alpha \text{ length}$ again	
 $R \triangleq \text{length} \leq \text{length}^\circ \triangleq \text{edit}$	

$$QF(\text{edit}^\circ)[\text{nil}, \text{cons}] \sqsubseteq F(\text{edit}^\circ)[\text{nil}, \text{cons}]R$$

every edit sequence built from a decomposition Q puts above a given one — any operation, each string lengthened at the front — is at least as long as one built from the given one

(15.2d)

circuit	Hinze–Marsden
$Q \triangleq 1 + (U \times V) \text{ is } F(U, V) \text{ at } U \triangleq \tau \text{ — } \triangleq \text{edit}; \text{ and the bifunctor } F \text{ preserves composition: } F(U, V)F(1, \text{edit}^\circ) = F(U, V \text{ edit}^\circ)$	
$F(T, \text{edit}^\circ R) \text{ — } (15.2e)$	
$F \text{ preserves composition: } F(U, \text{edit}^\circ R) = F(1, \text{edit}^\circ)F(U, R)$	
$F[Char]^2 \text{ — } R \text{ — } [Op(Char)]$	

edit $V^\circ \sqsubseteq R^\circ \text{edit}$ cutting a front off either string an edit sequence produces (edit V°) leaves a pair that a sequence no longer produces (R° edit)	
circuit	Hinze–Marsden
$V \triangleq \text{suffix}^\circ \times \text{suffix}^\circ, \text{ and } \times \text{ preserves composition}$	
$\text{edit} (\text{suffix} \times 1) \sqsubseteq R^\circ \text{edit: drop the operation that produced the head, or weaken its cpy to an ins. Proved by induction at suffix directly, not through suffix=tail* and } BA \sqsubseteq CB \implies BA^* \sqsubseteq C^*B \text{ as the book does}$	

circuit	Hinze–Marsden
$\text{edit } (1 \times \text{suffix}) \subseteq R^\circ \text{ edit}$, the mirror image: <code>cpy</code> weakens to a <code>del</code>	
R° is transitive, $\leq$ being so	

step $\text{base}^\circ = \emptyset$ no pair of strings is both a result of step and the result $([], [])$ of base		(15.2f)
circuit	Hinze–Marsden	
base returns only $([], [])$, so $\text{base}^\circ = \text{empty } \text{base}^\circ$		
$\text{step } \text{empty} = \emptyset$: <code>cpy</code> and <code>del</code> put a character on the left string, <code>ins</code> one on the right		
$\emptyset$ composed with anything is $\emptyset$		

$\text{mle}(xs, ys) = \text{head}(\text{column}(xs, ys))$, $\text{column}(xs, ys) = [\text{mle}(u, ys) \mid u \leftarrow \text{tails}(xs)]$ $\text{column}(xs) = ([\text{fstcol}(xs), \text{nextcol}(xs)] \Downarrow)$, $\text{fstcol} = \text{list}(\text{del}) \text{ tails}$ the tabulation: $\text{mle}(xs, ys)$ needs $\text{mle}(u, v)$ for every tail u of xs and v of ys , so the columns are built right to left $\text{column}(xs, \text{cons}(b, ys)) = \text{nextcol}(xs, (b, \text{column}(xs, ys)))$ $\text{nextcol}(xs)(b, us) = ([\text{base}(b, \text{last}(us)), \text{step}(b)] \Downarrow)(xus)$, $xus = \text{zip}(xs, \text{zip}(\text{init}(us), \text{tail}(us)))$ each column is a fold built bottom to top, over xs zipped with the adjacent pairs of the column to its right $\text{base}(b, u) = [[\text{ins}(b)] \# u]$ $\text{step}(b)((a, (u, v)), ws) = (a \rightarrow b \rightarrow [[\text{cpy}(a)] \# v] \# ws, [\text{bmin}(R)([\text{del}(a)] \# w, [\text{ins}(b)] \# u)] \# ws)$ $w = \text{head}(ws)$; these base , step are not edit 's. An entry depends on the one below it (a delete), the one to its right (an insert), and the one below that (a copy) — quadratic in the two lengths	(15.2g)
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§15.3. Optimal bracketing

tree $A ::= \text{tip } A \mid \text{bin } (\text{tree } A, \text{tree } A)$, $FX = A + X^2$, so $F(R) = 1 + R^2$; $h \triangleq [\text{tip}, \text{bin}]$, $\text{flatten} \triangleq ([\text{wrap}, \text{cat}] \Downarrow) : \text{tree } A \rightarrow \text{list}^+ A$, $H = \text{flatten}^\circ$. $\langle \text{cost}, \text{size} \rangle \triangleq ([\text{opt}, \text{opb}] \Downarrow)$, $\text{opt} \triangleq (\text{zero}, \text{st})$, $\text{opb } ((cx, sx), (cy, sy)) = (\text{cb } (sx, sy) + cx + cy, \text{sb } (sx, sy))$. sb associative, so $\text{size} = \text{flatten } sz$ for a map sz $R \triangleq \text{cost} \leq \text{cost}^\circ$, $g \triangleq [\text{zero}, (1 \times sz)^2 \text{ opb } \pi_1]$, single the coreflexive on singleton lists.	(15.3a)
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$\text{splits} \triangleq \langle \text{inits}^+, \text{tails}^+ \rangle$ zip, an implementation of $\frac{\text{cat}^0}{\text{}}_3$; $\text{array} \triangleq \text{inits list}(\text{row})$, $\text{row} \triangleq \text{tails list}(\text{mct})$, $\text{col} \triangleq \text{inits list}(\text{mct})$.

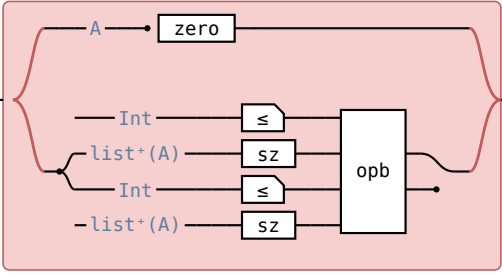
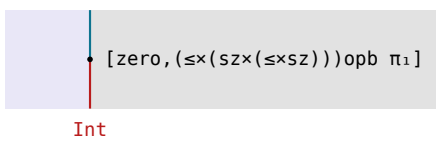
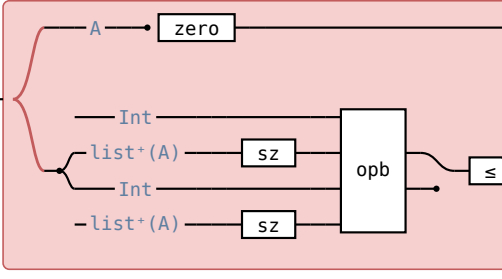
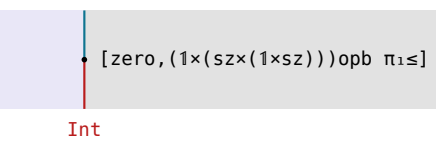
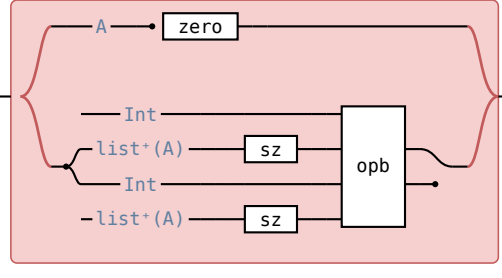
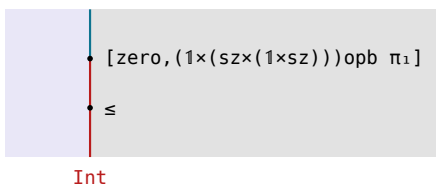
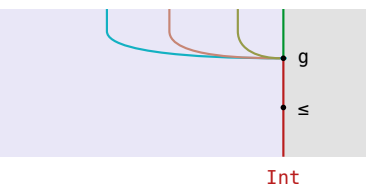
$\text{mix} \triangleq \text{zip list}(\text{bin}) \text{minlist}(\text{R})$, $\text{next} \triangleq \langle \pi_1, \text{mix} \rangle$ snoc , $\text{process} \triangleq ((\text{tip wrap}) \times 1) \text{ loop}(\text{next})$.

(15.3b)

$\text{Assoc}(\text{sb}) \Rightarrow \mathbf{F}(\langle \text{cost}, \text{flatten} \rangle) \mathbf{g} = [\text{tip}, \text{bin}] \text{cost}$ pairing each subtree with its cost and its flattening and then applying $\mathbf{g}$ gives what building the tree by $[\text{tip}, \text{bin}]$ and taking its cost gives	
circuit	Hinze–Marsden
$\mathbf{F}(\text{tree}(A)) \rightarrow \boxed{\mathbf{F}(\langle \text{cost}, \text{flatten} \rangle)} \rightarrow \mathbf{F}(\text{Int} \times \text{list}^+(A)) \rightarrow \boxed{\mathbf{g}} \rightarrow \text{Int}$	
$A + \text{tree}(A)^2 \rightarrow \left\{ \begin{array}{l} A \rightarrow \text{zero} \\ \text{tree}(A) \rightarrow \text{cost} \\ \text{tree}(A) \rightarrow \text{flatten} \rightarrow \text{list}^+(A) \rightarrow \text{sz} \end{array} \right\} \rightarrow \text{opb} \rightarrow \text{Int}$ definition of $\mathbf{g}$; coproducts and products $\text{—} \triangleq \text{flatten}$	$A \oplus \text{tree } A \times \text{tree } A \rightarrow [\text{zero}, ((\text{cost}, \text{flatten } \text{sz}) \times (\text{cost}, \text{flatten } \text{sz})) \text{opb } \pi_1] \rightarrow \text{Int}$
$A + \text{tree}(A)^2 \rightarrow \left\{ \begin{array}{l} A \rightarrow \text{zero} \\ \text{tree}(A) \rightarrow \text{cost} \\ \text{tree}(A) \rightarrow \text{size} \end{array} \right\} \rightarrow \text{opb} \rightarrow \text{Int}$ $\text{flatten } \text{sz} = \text{size}$, since sb is associative $\text{—} \triangleq \text{flatten}$	$A \oplus \text{tree } A \times \text{tree } A \rightarrow [\text{zero}, ((\text{cost}, \text{size}) \times (\text{cost}, \text{size})) \text{opb } \pi_1] \rightarrow \text{Int}$
$A + \text{tree}(A)^2 \rightarrow \left\{ \begin{array}{l} A \rightarrow \text{zero} \\ \text{tree}(A) \rightarrow \text{bin} \rightarrow \text{cost} \\ \text{tree}(A) \rightarrow \text{bin} \rightarrow \text{size} \end{array} \right\} \rightarrow \text{opb} \rightarrow \text{Int}$ $\langle \text{cost}, \text{size} \rangle \triangleq \langle \text{opt}, \text{opb} \rangle$, at a node $\text{bin}(\text{cost}, \text{size}) = (\text{cost}, \text{size})^2 \text{ opb}$	$A \oplus \text{tree } A \times \text{tree } A \rightarrow [\text{zero}, \text{bin}(\text{cost}, \text{size}) \pi_1] \rightarrow \text{Int}$
$A + \text{tree}(A)^2 \rightarrow \left\{ \begin{array}{l} A \rightarrow \text{tip} \rightarrow \text{cost} \\ \text{tree}(A) \rightarrow \text{bin} \rightarrow \text{cost} \end{array} \right\} \rightarrow \text{opb} \rightarrow \text{Int}$ $\text{tip } \text{cost} = \text{zero}$; products, $\langle \text{cost}, \text{size} \rangle \pi_1 = \text{cost}$	$A \oplus \text{tree } A \times \text{tree } A \rightarrow [\text{tip } \text{cost}, \text{bin } \text{cost}] \rightarrow \text{Int}$
$A + \text{tree}(A)^2 \rightarrow \left\{ \begin{array}{l} A \rightarrow \text{tip} \\ \text{tree}(A) \rightarrow \text{bin} \end{array} \right\} \rightarrow \text{tree}(A) \rightarrow \text{cost} \rightarrow \text{Int}$ coproducts, $[\text{tip } \text{cost}, \text{bin } \text{cost}] = [\text{tip}, \text{bin}] \text{ cost}$	$A \oplus \text{tree } A \times \text{tree } A \rightarrow \text{tree} \cdot [\text{tip}, \text{bin}] \rightarrow \text{cost} \rightarrow \text{Int}$

(15.3c)

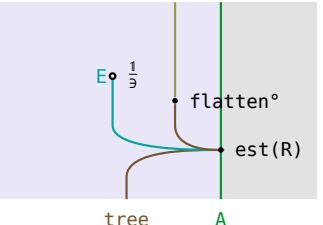
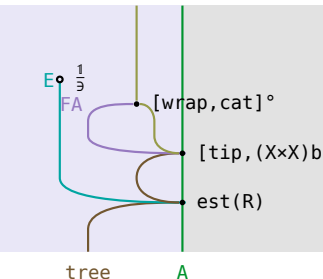
$\mathbf{F}(\leq \times 1) \mathbf{g} \mathbf{E} \mathbf{g} \leq$ raising the cost of either subtree ($\leq \times 1$ in both slots) and then applying $\mathbf{g}$ returns a value at least what $\mathbf{g}$ returns on the costs before raising ($\mathbf{g} \leq$)	
circuit	Hinze–Marsden
$\mathbf{F}(\text{Int} \times \text{list}^+(A)) \rightarrow \boxed{\mathbf{F}(\leq \times 1)} \rightarrow \boxed{\mathbf{g}} \rightarrow \text{Int}$	

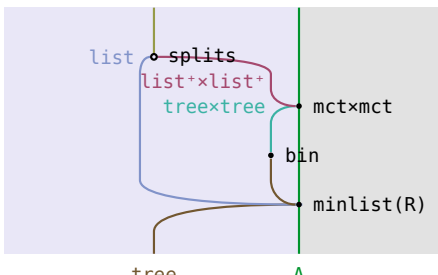
circuit	Hinze–Marsden
$A + \text{Int} \times \text{list}^+(A) \times \text{Int} \times \text{list}^+(A)$  definition of $g \triangleq \text{flatten}$	$A \oplus (\text{Int} \times \text{list}^+ A) \times \text{Int} \times \text{list}^+ A$ 
$A + \text{Int} \times \text{list}^+(A) \times \text{Int} \times \text{list}^+(A)$  definition of opb, and $+$ monotonic	$A \oplus (\text{Int} \times \text{list}^+ A) \times \text{Int} \times \text{list}^+ A$ 
$A + \text{Int} \times \text{list}^+(A) \times \text{Int} \times \text{list}^+(A)$  $\leq$ reflexive, so $\text{zero} \leq \text{zero}$; coproducts	$A \oplus (\text{Int} \times \text{list}^+ A) \times \text{Int} \times \text{list}^+ A$ 
$F(\text{Int} \times \text{list}^+(A)) \xrightarrow{g} \text{Int}$ definition of $g \triangleq \text{flatten}$	

$\text{Assoc}(\text{sb}) \Rightarrow (\mu X : \frac{[\text{wrap}, \text{cat}]^\circ}{\exists} P([\text{tip}, (X \times X) \text{bin}]) \text{est}(R)) \stackrel{\text{flatten}^\circ}{\exists} \text{est}(R)$, $\text{mct} = (\text{single-head tip}, (\text{init col}, \text{tail row}) \text{mix})$

a least-cost bracketing of $a_1 \oplus \dots \oplus a_n$ is read off an array holding one best tree per non-empty segment, each entry built from the column to its left and the row below it

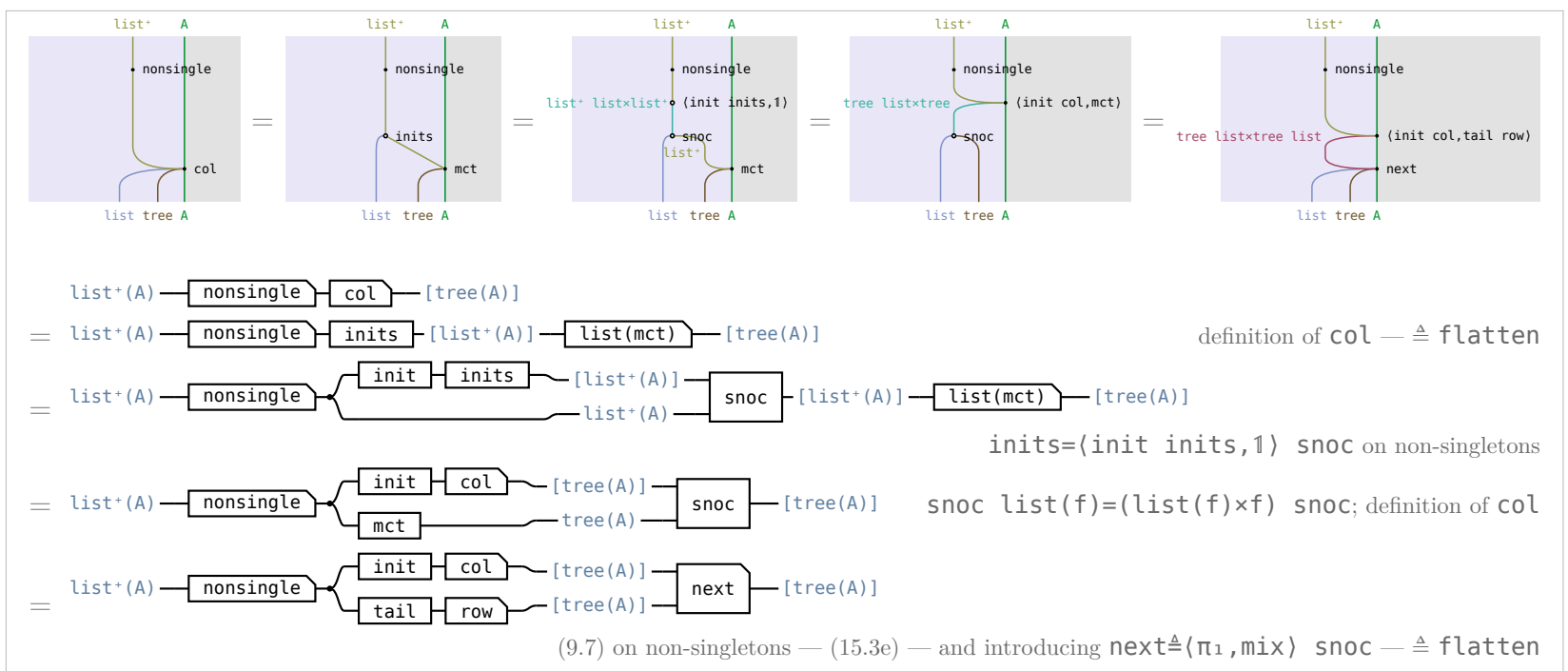
(15.3d)

circuit	Hinze–Marsden
$\text{list}^+(A) \xrightarrow{\frac{1}{\exists}} E(\text{list}^+(A)) \xrightarrow{E(\text{flatten}^\circ)} E(\text{tree}(A)) \xrightarrow{\text{est}(R)} \text{tree}(A)$ the specification $\triangleq \text{flatten}$	
$\text{list}^+(A) \xrightarrow{\frac{1}{\exists}} E(\text{list}^+(A)) \xrightarrow{E([\text{wrap}, \text{cat}]^\circ)} E(\text{list}^+(A)) \xrightarrow{P([\text{tip}, (X \times X) \text{bin}])} E(\text{tree}(A)) \xrightarrow{\text{est}(R)} \text{tree}(A)$ split the list in every way, bracket both halves, join. The condition is monotonicity in context, $F(Rn(\text{flatten} \text{ flatten}^\circ)) \text{h} \leq \text{h} R$ — only trees with the same flattening are compared — which is Proposition 9.3 at $H^\circ = \text{flatten}$, a map, with (9.5) $[\text{tip}, \text{bin}]$ $\text{cost} = (1 + \langle \text{cost}, \text{flatten} \rangle^2) g$ (the cost of a node reads only the cost and the flattening of its two subtrees) and (9.6) $(1 + (\leq 1)^2) g \leq g$ (g monotonic on $\leq$ in its two cost arguments)	
$\text{list}^+(A) \xrightarrow{\frac{1}{\exists}} E(\text{list}^+(A)) \xrightarrow{E(\text{cat}^\circ)} E(\text{list}^+(A)^2) \xrightarrow{P((X \times X) \text{bin})} E(\text{tree}(A)) \xrightarrow{\text{est}(R)} \text{tree}(A)$ Proposition 9.1: wrap and cat have disjoint ranges, and single is the coreflexive on singleton lists, where wrap returns	

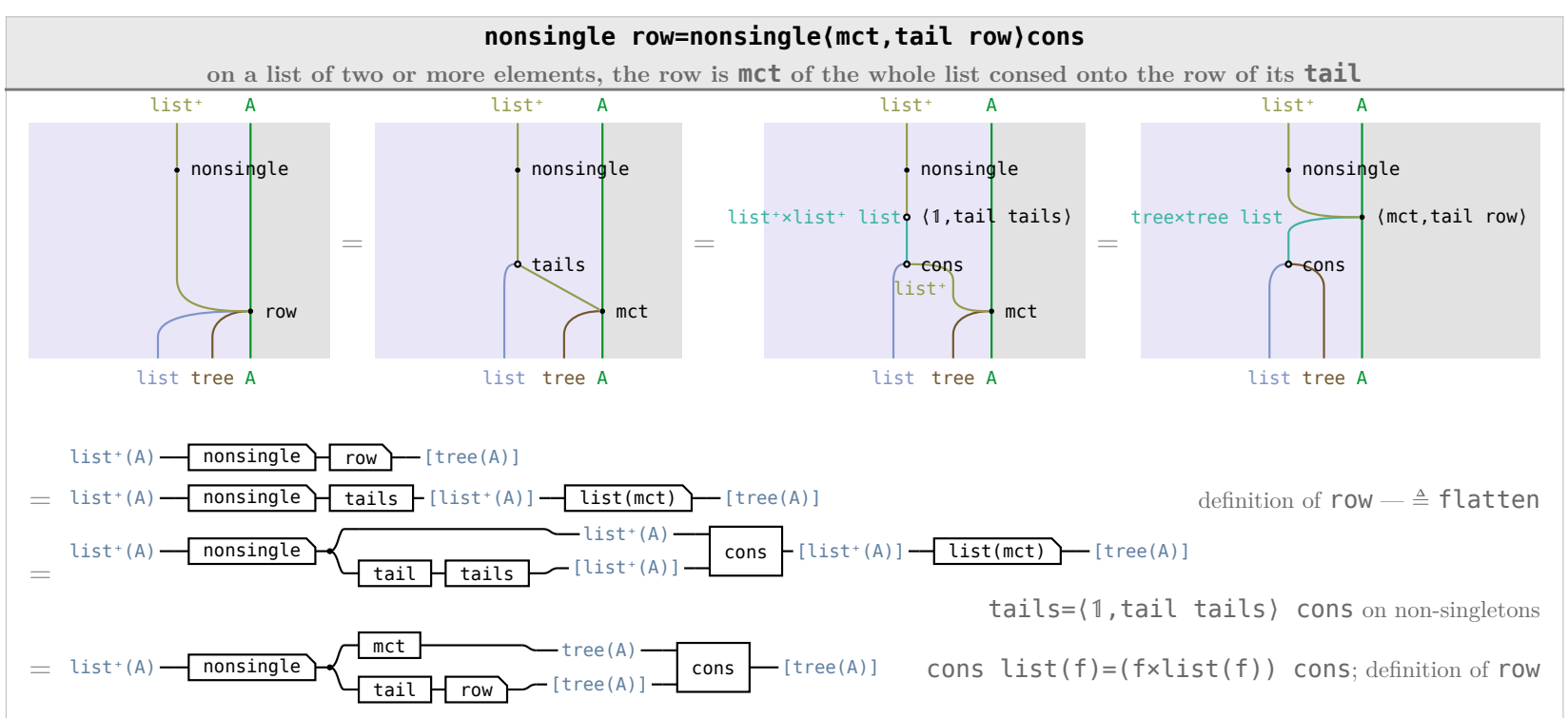
circuit	Hinze–Marsden
$\text{list}^+(A) \rightarrow \boxed{\text{splits}} \rightarrow [\text{list}^+(A)^2] \rightarrow \boxed{\text{list}((\text{mct} \times \text{mct}) \text{bin})} \rightarrow [\text{tree}(A)] \rightarrow \boxed{\text{minlist}(R)} \rightarrow \text{tree}(A)$ $\sqsubseteq$ $\text{splits} \triangleq \langle \text{inits}^+, \text{tails}^+ \rangle$ zip implements $\frac{\text{cat}}{\circ}$ and $\text{minlist}(R)$ implements $\text{est}(R)$. Exponential, since the segments of one list overlap	
$\text{mct} = (\text{single} \rightarrow \text{head tip}, \langle \text{init col}, \text{tail row} \rangle \text{mix}) \quad (9.7)$ $=$ the tabulation: $\text{mct } xS$ is needed for every non-empty segment xS, so the values are held as an array of rows, $\text{array} \triangleq \text{inits } \text{list}(\text{row}), \text{row} \triangleq \text{tails } \text{list}(\text{mct}), \text{col} \triangleq \text{inits } \text{list}(\text{mct}), \text{mix} \triangleq \text{zip } \text{list}(\text{bin}) \text{ minlist}(R)$	
$\text{col} = (\text{single} \rightarrow \text{head tip wrap}, \langle \text{init col}, \text{tail row} \rangle \text{next}) \quad (9.8)$ $\text{cons col} = (1 \times \text{array}) \text{ process} \quad (9.9)$ $=$ $\text{row} = (\text{single} \rightarrow \text{head tip wrap}, \langle \text{mct}, \text{tail row} \rangle \text{cons}) \quad (9.10)$ a column extends the column to its left, a row the row below it; (9.9) is (9.8) rewritten as a loop, $\text{next} \triangleq \langle \pi_1, \text{mix} \rangle \text{ snoc}, \text{process} \triangleq ((\text{tip wrap}) \times 1) \text{ loop}(\text{next})$	
$\text{array} = ([\text{fstcol}, \text{addcol}]), \text{fstcol} \triangleq \text{tip wrap wrap}$ $=$ $\text{addcol} \triangleq \langle \pi_1 \text{ tip wrap}, \text{step} \rangle \text{ cons}, \text{step} \triangleq (\text{process tail}, \pi_2) \text{ zip } \text{list}(\text{cons})$ the program: one fold building the array column by column, cubic in the length of the input	

nonsingle mct = nonsingle(init col, tail row)mix on a list of two or more elements, mct gives what mix gives on the column of its init and the row of its tail	
	
$\text{list}^+(A) \rightarrow \boxed{\text{nonsingle}} \rightarrow \boxed{\text{mct}} \rightarrow \text{tree}(A)$ $=$ $\text{list}^+(A) \rightarrow \boxed{\text{nonsingle}} \rightarrow \boxed{\text{splits}} \rightarrow [\text{list}^+(A)^2] \rightarrow \boxed{\text{list}(\text{graft})} \rightarrow [\text{tree}(A)] \rightarrow \boxed{\text{minlist}(R)} \rightarrow \text{tree}(A)$ recursive case of mct — $\triangleq$ flatten $=$ $\text{list}^+(A) \rightarrow \boxed{\text{nonsingle}} \rightarrow \left\{ \begin{array}{l} \text{inits}^+ \\ \text{tails}^+ \end{array} \right\} \rightarrow [\text{list}^+(A)] \rightarrow \boxed{\text{zip}} \rightarrow [\text{list}^+(A)^2] \rightarrow \boxed{\text{list}(\text{graft})} \rightarrow [\text{tree}(A)] \rightarrow \boxed{\text{minlist}(R)} \rightarrow \text{tree}(A)$ definition of splits — $\triangleq$ flatten $=$ $\text{list}^+(A) \rightarrow \boxed{\text{nonsingle}} \rightarrow \boxed{\text{trees}} \rightarrow \boxed{\text{zip}} \rightarrow [\text{tree}(A)^2] \rightarrow \boxed{\text{list}(\text{bin})} \rightarrow [\text{tree}(A)] \rightarrow \boxed{\text{minlist}(R)} \rightarrow \text{tree}(A)$ definitions of $\text{graft} \triangleq (\text{mct} \times \text{mct})$ bin and $\text{trees} \triangleq (\text{inits}^+ \text{ list}(\text{mct}), \text{tails}^+ \text{ list}(\text{mct}))$, our names; $\text{zip } \text{list}(f \times g) = (\text{list}(f) \times \text{list}(g)) \text{ zip}$ $=$ $\text{list}^+(A) \rightarrow \boxed{\text{nonsingle}} \rightarrow \boxed{\text{trees}} \rightarrow \boxed{\text{mix}} \rightarrow \text{tree}(A)$ introducing $\text{mix} \triangleq \text{zip } \text{list}(\text{bin}) \text{ minlist}(R)$ — $\triangleq$ flatten $=$ $\text{list}^+(A) \rightarrow \boxed{\text{nonsingle}} \rightarrow \left\{ \begin{array}{l} \text{init} \quad \text{col} \\ \text{tail} \quad \text{row} \end{array} \right\} \rightarrow [\text{tree}(A)] \rightarrow \boxed{\text{mix}} \rightarrow \text{tree}(A)$ definition of trees; $\text{inits}^+ = \text{init } \text{inits}$ and $\text{tails}^+ = \text{tail } \text{tails}$ on non-singletons; definition of row and col — $\triangleq$ flatten	

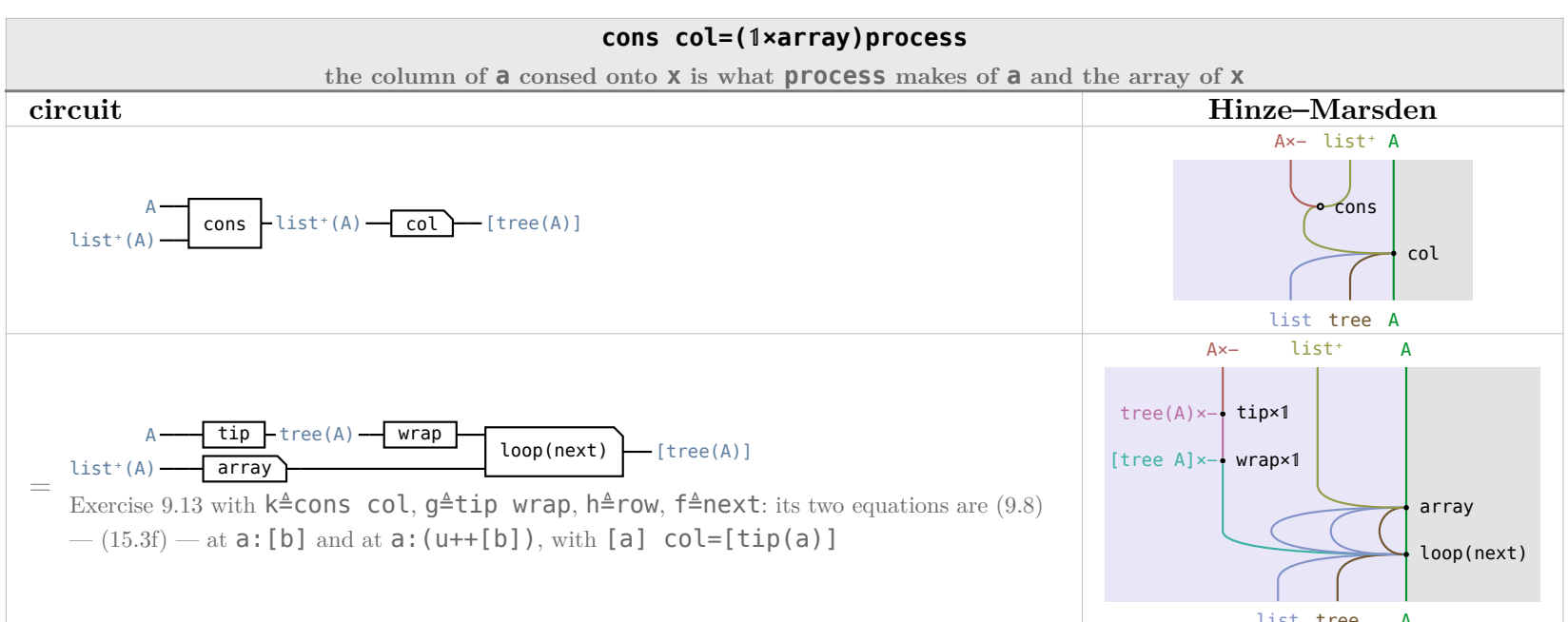
nonsingle col = nonsingle(init col, tail row)next on a list of two or more elements, the column is what next makes of the column of its init and the row of its tail	
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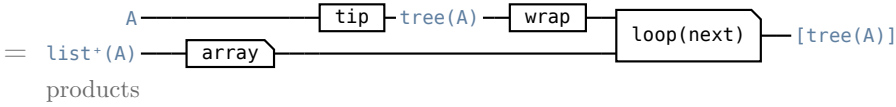
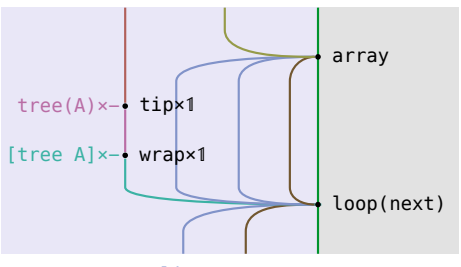
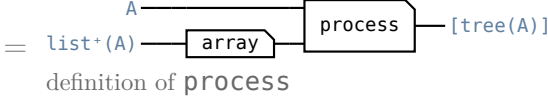
(15.3g)



(15.3h)



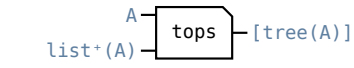
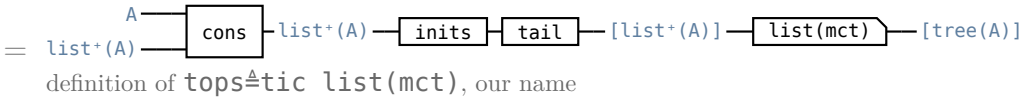
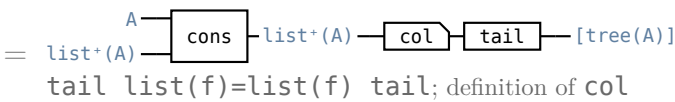
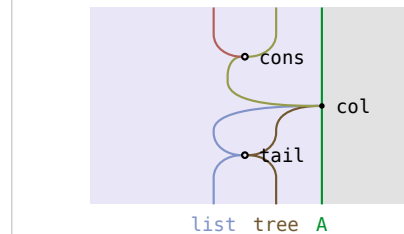
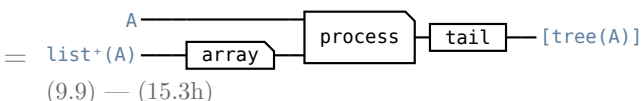
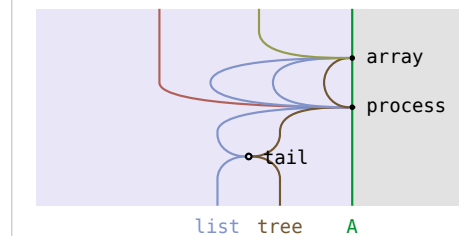
(15.3h)

circuit	Hinze–Marsden
	
	

tops=(1×array)process tail

the trees at the top of the new rows are what **process** makes of **a** and the array of **x**, less its first

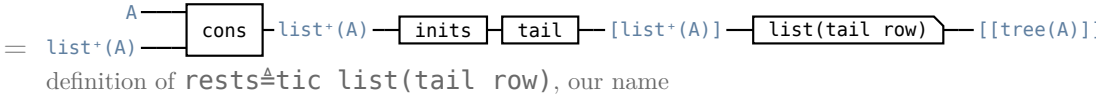
(15.3i)

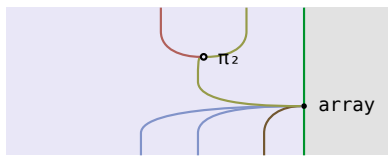
circuit	Hinze–Marsden
	
	
	
	

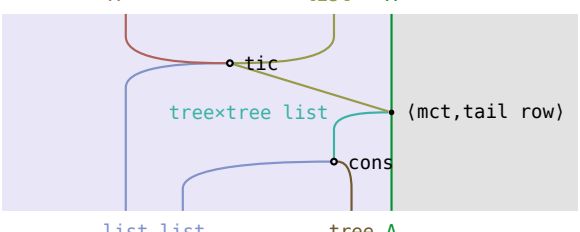
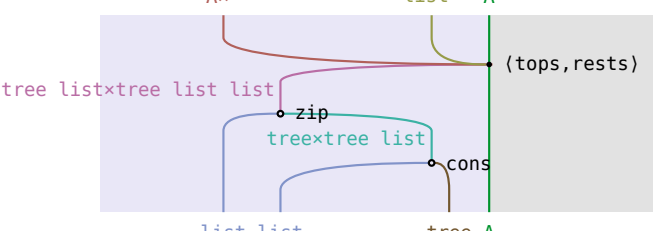
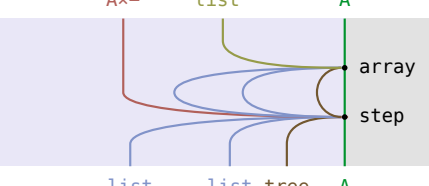
rests=π₂ array

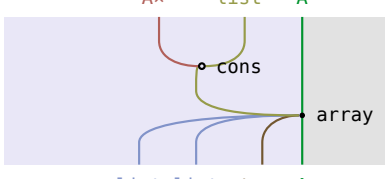
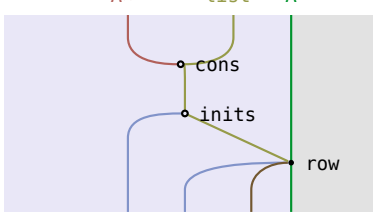
the rest of each new row is the array of **x**

(15.3j)

circuit	Hinze–Marsden
	
	

circuit	Hinze–Marsden
$= \text{list}^+(A) \xrightarrow{\text{array}} [[\text{tree}(A)]]$ $\text{tic list}(\text{tail}) = \pi_2$ inits; definition of array	

newrows=(1×array)step the rows of the array of a:x below its first are what step makes of a and the array of x		(15.3k)
circuit	Hinze–Marsden	
$\text{list}^+(A) \xrightarrow{\text{newrows}} [[\text{tree}(A)]]$		
$= \text{list}^+(A) \xrightarrow{\text{cons}} \text{list}^+(A) \xrightarrow{\text{inits}} \text{tail} \xrightarrow{[\text{list}^+(A)]} \text{list}(\text{row}) \xrightarrow{[[\text{tree}(A)]]}$ definition of newrows $\hat{=}$ tic list(row), our name		
$= \text{list}^+(A) \xrightarrow{\text{cons}} \text{list}^+(A) \xrightarrow{\text{inits}} \text{tail} \xrightarrow{[\text{list}^+(A)]} \text{list}(\text{mct, tail row cons}) \xrightarrow{[[\text{tree}(A)]]}$ (9.10) — (15.3g) — on the non-singleton tic lists		
$= \text{list}^+(A) \xrightarrow{\text{tops}} \text{rests} \xrightarrow{\text{zip}} [\text{tree}(A) \times \text{tree}(A)] \xrightarrow{\text{list(cons)}} [[\text{tree}(A)]]$ $\text{list}(f,g) = (\text{list}(f), \text{list}(g))$ zip; products; definitions of tops and rests		
$= \text{list}^+(A) \xrightarrow{\text{array}} \text{step} \xrightarrow{[[\text{tree}(A)]]}$ (15.3i) and (15.3j); products; definition of step		

cons array=(1×array)addcol the array of a consed onto x is what addcol makes of a and the array of x		(15.3l)
circuit	Hinze–Marsden	
$\text{list}^+(A) \xrightarrow{\text{cons}} \text{list}^+(A) \xrightarrow{\text{array}} [[\text{tree}(A)]]$		
$= \text{list}^+(A) \xrightarrow{\text{cons}} \text{list}^+(A) \xrightarrow{\text{inits}} [\text{list}^+(A)] \xrightarrow{\text{list(row)}} [[\text{tree}(A)]]$ definition of array		

circuit	Hinze–Marsden
$\text{cons inits} = (\pi_1 \text{ wrap}, \text{tic}) \text{ cons}$, abbreviating cons inits tail by tic	
$\text{cons list}(f) = (f \times \text{list}(f)) \text{ cons}$; definition of newrows	
$\text{wrap row} = \text{tip wrap}$	
$\text{wrap row} = \text{tip wrap}$	

§15.4. Data compression

$[A] ::= \text{nil} \mid \text{snoc } ([A], A), \text{ list}^+ A ::= \text{wrap } A \mid \text{snoc } (\text{list}^+ A, A), \text{ String} = [\text{Char}], \text{ Code} ::= \text{sym Char} \mid \text{ptr} (\text{String}, \text{String}^+), F(\text{Code}, \text{String}) = 1 + (\text{String} \times \text{Code}), \alpha \triangleq [\text{nil}, \text{snoc}].$ (15.4a)

$\text{decode} \triangleq ([\text{nil}, \text{extend}]) : [\text{Code}] \rightarrow \text{String}, \text{ extend } (xs, \text{sym } a) = xs \# [a], \text{ extend } (xs, \text{ptr } (ys, zs)) = xs \# zs \text{ when } ys \# zs \text{ is a proper prefix of } xs \# zs; H = \text{decode}^\circ.$

$\text{size} \triangleq ([\text{zero}, \text{distr } [1 \times c, 1 \times p] \text{ plus}])$ with c, p the constant costs of a symbol and a pointer; $R \triangleq \text{size} \leq \text{size}^\circ$, $Q \triangleq F(\tau + \tau, \text{prefix}^\circ) = 1 + (\text{prefix}^\circ \times (\tau + \tau))$, the two τ on symbols and on pointers.

$\text{lrt } ws = \text{est}(\text{prefix}^\circ \times (\tau + \tau)) \{(xs, (ys, zs)) \mid xs \# zs = ws, ys \# zs \text{ a proper prefix of } ws\}$, the longest repeated tail; $\text{reduce } (ws \# [a]) = (zs \neq [] \rightarrow [(ws, \text{sym } a), (xs, \text{ptr } (ys, zs))], [(ws, \text{sym } a)])$ with $(xs, (ys, zs)) = \text{lrt } (ws \# [a])$.

$0 \leq c \wedge 0 \leq p \Rightarrow (\mu X : \frac{[\text{nil}, \text{extend}]^\circ}{3} \text{ thin}(Q) P([\text{nil}, (X \times 1) \text{snoc}]) \text{ est}(R)) \sqsubseteq \frac{\text{decode}^\circ}{3} \text{ est}(R), \text{ encode} = (\text{null} \rightarrow \text{nil}, \text{reduce list}((\text{encode} \times 1) \text{snoc}) \text{ minlist}(R))$ (15.4b)

a smallest code sequence decoding to the given string is built from the right, each step emitting the last character as a symbol or ending with a pointer back into what has already been decoded

circuit	Hinze–Marsden
the specification $\triangleq \text{decode}$	
at $Q \triangleq F(\tau + \tau, \text{prefix}^\circ) = 1 + (\text{prefix}^\circ \times (\tau + \tau))$, the two τ on symbols and on pointers. Monotonicity $F(R) \alpha \sqsubseteq \alpha R$ is routine, the two costs being constants: $(\tau + \tau)[c, p] = [c, p]$. $\square$ Proposition 9.4 splits the thinning condition in two — $F(\tau + \tau, R) \alpha \sqsubseteq \alpha R$, and $\text{decode prefix} \sqsubseteq R^\circ \text{ decode}$, for which $\text{decode init} \sqsubseteq R^\circ \text{ decode}$ is enough: dropping the last character of the output shortens or removes the last code element and never raises the cost. So between a symbol and a pointer nothing can be decided in advance, and between two pointers the longer match wins	

circuit	Hinze–Marsden
$ \begin{aligned} & [\text{Char}] \xrightarrow{\frac{1}{5}} E[\text{Char}] \xrightarrow{E(\text{extend}^\circ)} \text{thin}(\text{prefix}^\circ \times (\text{T}+\text{T})) \xrightarrow{E([\text{Char}] \times \text{Code})} P((X \times 1) \text{snoc}) \xrightarrow{E[\text{Code}]} \text{est}(R) \xrightarrow{[\text{Code}]} \\ = & \text{Proposition 9.1: } \text{nil} \text{ and } \text{extend} \text{ have disjoint ranges. The decompositions of one string} \\ & \text{are } \frac{\text{extend}^\circ}{\supseteq} (\text{ws} \# [\text{a}]) = \{(\text{ws}, \text{sym } \text{a})\} \cup \{(\text{xs}, \text{ptr } (\text{ys}, \text{zs})) \mid \text{xs} \# \text{zs} = \text{ws} \# [\text{a}], \\ & \text{ys} \# \text{zs} \text{ a proper prefix of } \text{ws}\} \text{ — take the last character as a symbol, or end with a pointer} \end{aligned} $	
$ \begin{aligned} & [\text{Char}] \xrightarrow{\text{reduce}} [[\text{Char}] \times \text{Code}] \xrightarrow{\text{list}((\text{encode} \times 1) \text{snoc})} [[\text{Code}]] \xrightarrow{\text{minlist}(R)} [\text{Code}] \\ = & \text{reduce implements } \frac{\text{extend}^\circ}{\supseteq} \text{thin}(\text{prefix}^\circ \times (\text{T}+\text{T})): \text{thinning leaves at most two, the} \\ & \text{symbol and the pointer of the longest repeated tail } \text{lrt}. \text{ Again exponential; the book gives} \\ & \text{no tabulation for it} \end{aligned} $	

$$0 \leq c \wedge 0 \leq p \Rightarrow QF(\text{decode}^\circ)[\text{nil}, \text{snoc}] \sqsubseteq F(\text{decode}^\circ)[\text{nil}, \text{snoc}]R$$

(15.4c)

every code sequence built from a decomposition Q puts above a given one — a code element of the same kind, a longer front string — costs at least as much as one built from the given one

circuit	Hinze–Marsden
$ \begin{aligned} & [\text{Char}] \xrightarrow{\text{Code}} F((\text{T}+\text{T}), \text{prefix}^\circ) \xrightarrow{\text{Code}} \text{nil} \xrightarrow{\text{Code}} [\text{Code}] \\ & \text{Q} \triangleq 1 + (\text{prefix}^\circ \times (\text{T}+\text{T})) \text{ is } F(\text{T}+\text{T}, \text{prefix}^\circ) \text{ — } \triangleq \text{decode} \end{aligned} $	
$ \begin{aligned} & [\text{Char}] \xrightarrow{\text{Code}} F((\text{T}+\text{T}), \text{prefix}^\circ \text{decode}^\circ) \xrightarrow{\text{Code}} \text{nil} \xrightarrow{\text{Code}} [\text{Code}] \\ & \text{the bifunctor } F \text{ preserves composition: } F(U, \text{prefix}^\circ)F(1, \text{decode}^\circ) = F(U, \text{prefix}^\circ \text{decode}^\circ) \end{aligned} $	
$ \begin{aligned} & [\text{Char}] \xrightarrow{\text{Code}} F((\text{T}+\text{T}), \text{decode}^\circ R) \xrightarrow{\text{Code}} \text{nil} \xrightarrow{\text{Code}} [\text{Code}] \\ & \text{Proposition 9.4's second condition } \text{prefix}^\circ \text{decode}^\circ \sqsubseteq \text{decode}^\circ R: \text{dropping the last} \\ & \text{character drops the last } \text{sym}, \text{ or shortens or drops the last pointer. Proved at } \text{prefix} \\ & \text{directly by induction, not through } \text{init} \end{aligned} $	
$ \begin{aligned} & [\text{Char}] \xrightarrow{\text{Code}} F(\text{decode}^\circ) \xrightarrow{\text{Code}} F((\text{T}+\text{T}), R) \xrightarrow{\text{Code}} \text{nil} \xrightarrow{\text{Code}} [\text{Code}] \\ & F \text{ preserves composition: } F(U, \text{decode}^\circ R) = F(1, \text{decode}^\circ)F(U, R) \end{aligned} $	

circuit	Hinze–Marsden
Proposition 9.4's first condition $F(T+T, R) \alpha \alpha R$, left as an exercise in the book: <code>snoc</code> adds the cost of the last element to both sides, and $[c, p] (T+T)=[c, p]$	

extend nil°=⊘ no string is both a result of extend and the result [] of nil	
circuit	Hinze–Marsden
nil returns only <code>[]</code>, so $\text{nil}^\circ = \text{null } \text{nil}^\circ$	
extend null=⊘: a symbol ends the string with a character, a pointer with its non-empty <code>zs</code>	
$\emptyset$ composed with anything is $\emptyset$	

(15.4d)